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Timing and Tempo in Development

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Miki Ebisuya is a group leader at Physics of Life TU Dresden/EMBL Barcelona. Her laboratory recapitulates species-specific biological time *in vitro* using the stem cell zoo, a collection of pluripotent stem cells from diverse animal species. Her laboratory also seeks to manipulate the species-specific time.

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James Briscoe is a group leader at the Francis Crick Institute, London. His laboratory's research interests include the molecular and cellular mechanisms of tissue development and the role of transcriptional networks in the specification of cell fate. To address these questions, his laboratory uses a range of experimental and computational techniques with model systems that include mouse, human and chick embryos and embryonic stem cells.

In this issue of Current Opinion in Genetics and Development, we bring together a series of review articles on the theme of 'Developmental Timing'. These articles address a fundamental issue at the heart of developmental biology. From determining the sequence of events in embryo development to balancing growth and differentiation, the precise timing and tempo of molecular and cellular processes are crucial to the formation and function of tissues and organs. Altering the pace of development can affect the size and makeup of tissues, as well as their function. Imbalances in the rate of tissue growth and differentiation can cause overgrowth or morphological defects. Moreover, although the sequence and underlying molecular mechanisms controlling biological processes are often similar across species, the tempo at which they progress can vary substantially. Comparisons between species are beginning to offer insight into timekeeping mechanisms and how they evolved.

The themes covered in the articles include discussions of the mechanisms, functions, and evolution of timing in developmental processes. The articles aim to shed light on how the timing and pace of developmental events are regulated and how these factors influence the formation and function of biological structures. There are insights from various disciplines, including developmental and cell biology, evolutionary biology, physiology, and physics; together, these aim to provide a comprehensive overview of our current understanding of developmental timing. The articles offer perspectives on the similarities and differences in how these processes are controlled and have evolved. By integrating insights from various fields, we hope the volume advances our understanding of this crucial aspect of life and its implications for developmental biology, evolutionary biology, and beyond.

Epigenetic and metabolic control

A fundamental aspect explored in several of the articles is the mechanisms that control developmental timing across a wide range of species. Of particular note is the recent interest in epigenetic and metabolic mechanisms.

Ciceri and Studer discuss recent findings that highlight the role for epigenetic factors in the timing of neuronal maturation. They summarise evidence that epigenetic regulators control post-translation modification of chromatin proteins to determine the pace of neuronal maturation in the cerebral cortex. This is consistent with recent theoretical work [1] that indicates that epigenetic mechanisms can readily and robustly alter the

speed of developmental timetables and might be a general strategy for controlling developmental tempo.

Rekaik and Duboule also make the case for the epigenetic control of gene expression timing. Hox genes are activated in a progressive rostral-to-caudal sequence during the elaboration of the bilateral body plan. Gradual changes in chromatin modification and structure of the Hox clusters have been implicated in sequential gene activation. Rekaik and Duboule describe how this results in the loading of cohesion complexes that initiate chromatin extrusion. In turn, extrusion is limited by the position and orientation of CCCTC-binding factor (CTCF) sites within the Hox locus, and this appears to set the pace for the activation of later Hox genes.

Metabolism has also been implicated in regulating developmental tempo. Casimir et al. discuss the evidence that mitochondrial metabolism contributes to the rate of neuronal maturation. As neurons mature, there are changes in mitochondria function. Accelerating this in human cortical neurons results in the accelerated maturation of the neurons; conversely, its reduction leads to decreased maturation rates in mouse neurons. Diaz-Cuadros also highlights the role of metabolism in controlling developmental tempo. A comparison of the human and mouse segmentation clock, the molecular oscillator involved in the rhythmic production of somites, revealed that the mouse clock cycles at two to three times the rate of the human clock. Diaz-Cuadros summarises evidence that the cytoplasmic NAD^+/NADH (Nicotinamide adenine dinucleotide) redox balance determines the rate of oscillations and contributes to the difference in the developmental tempo between mouse and human.

Metabolic and epigenetic control can collaborate to regulate developmental timing. Fenelon discusses how both mechanisms contribute to embryonic diapause, a period of developmental stasis that occurs in some species at the blastocyst stage. A transition from carbohydrate metabolism to lipid metabolism mediated by the FoxO1 transcription factor as well as multiple epigenetic modifications are essential to induce and maintain the diabase state in mammalian embryos.

These articles shed light on how metabolic and epigenetic status can influence developmental time in addition to genetic control.

Co-ordinating time in developing embryos

Several articles emphasise the absence of an external clock for regulating the tempo of development. Instead, internal control mechanisms allow developing embryos to co-ordinate tempo.

Morrow and Mirth discuss how timed pulses of the moulting hormone ecdysone are necessary for developmental progression in Drosophila fruit flies, guiding development through larval moults, pupation and metamorphosis. This allows the cell patterning and growth to be co-ordinated in a tissue-specific manner. Morrow and Mirth discuss the mechanisms of gene regulation involved in this process.

The circadian clock provides another means to co-ordinate biological function throughout the body. Yagita discusses the gradual emergence of circadian oscillation as cellular differentiation progresses and the oscillation of the segmentation clock weakens. This is recapitulated during *in vitro* differentiation of mouse pluripotent stem cells and organoids, indicating that the initiation mechanism of the circadian clock is cell autonomous and tightly coupled with other developmental timing mechanisms.

The nematode *Caenorhabditis elegans* has long served as a model system for studying the control of developmental timing. Patil and van Zon summarise studies that have identified the timers controlling larval development and the mechanisms that make timing robust to fluctuations in environmental conditions.

Collectively, these articles provide insight into the systematic mechanisms that allow long-range co-ordination of the rate of development.

Comparative studies and evolutionary perspectives

Comparative studies between species are emerging as an exciting means to gain insight into the mechanisms and evolution of developmental timekeeping.

Azzi and Rayon summarise recent findings and working hypotheses that underlie differential developmental tempo across mammalian species, with a specific focus on the pace of neural development and the segmentation clock. Both the extrinsic control of developmental tempo and the intrinsic roles of cell cycle, metabolic rates, protein turnover and post-transcriptional mechanisms are highlighted as potential mechanisms to give rise to species-specific developmental tempo.

Lazaro et al. also discuss how pluripotent stem cell–based models are being used to investigate temporal processes. A key advantage of stem cell–based models is the capacity to perform detailed quantitative comparisons under identical conditions, and the availability of stem cells from different species is extending these studies to a wide variety of species. With the establishment of organoid methods to study a range of tissues, this is opening the possibility of studying more diverse biological time, such as heart rate, diapause, lifespan and

the circadian clock. This is likely to offer insight into how differences in biological tempo between species can be a driver of evolutionary novelty.

Pigeon and Hassam also propose the existence of dedicated molecular mechanisms under evolutionary selection that control the tempo of neurogenesis. They propose a useful conceptual framework inspired by computer science to think about how these biological mechanisms operate during brain development and evolution. Together, these articles underscore the complexity of the mechanisms that control developmental tempo and emphasise that much work remains to be done before we can confidently identify the ultimate causes of species-specific developmental rates.

Developmental timing and tissue formation

In many tissues, temporal mechanisms affect cell type diversity and organisation of tissues. Sagner describes a recently identified temporal patterning programme that involves cascades of transcription factors that control the timing and identity of neurons generated in the vertebrate nervous system. Focusing on the developing spinal cord, he discusses how the mechanism contributes to the ordered formation of neuronal circuits.

Maeda and Kageyama summarise the role of ultradian oscillations, oscillatory gene expressions with shorter periods than the circadian rhythm, in tissue development. While ultradian oscillations have been extensively studied in the segmentation clock and neurogenesis, they also regulate and balance cell proliferation and differentiation processes in multiple tissues, including muscle and pancreatic development.

Sancho et al. describe the mechanisms by which plant roots are generated from specialist groups of cells at the root. This involves the co-ordination of cell division and the specification of cell identity for proper root development.

Bourn and Dorrity explore how temperature influences developmental tempo. Increasing the temperature accelerates development. Both proteostasis and metabolic rates are linked to temperature. Cutting-edge genomics methods are allowing the extraction of timing information from whole embryos at single-cell resolution. Bourn and Dorrity describe how this is beginning to reveal tissue-specific differences in the response and sensitivity to temperature changes.

In summary, this volume provides a multidisciplinary exploration of the mechanisms that regulate the timing and pace of development. By integrating insights from molecular biology, genetics, computational approaches, and evolutionary biology, we hope the articles contribute to our understanding of the fundamental principles underlying developmental timing and its implications for the formation and function of living organisms.

Reference

1. Nguyen P, Pease NA, Kueh HY: **Scalable control of developmental timetables by epigenetic switching networks**. *J R Soc Interface* (180) 2021, **18**:20210109, https://doi.org/10.1098/rsif.2021.0109